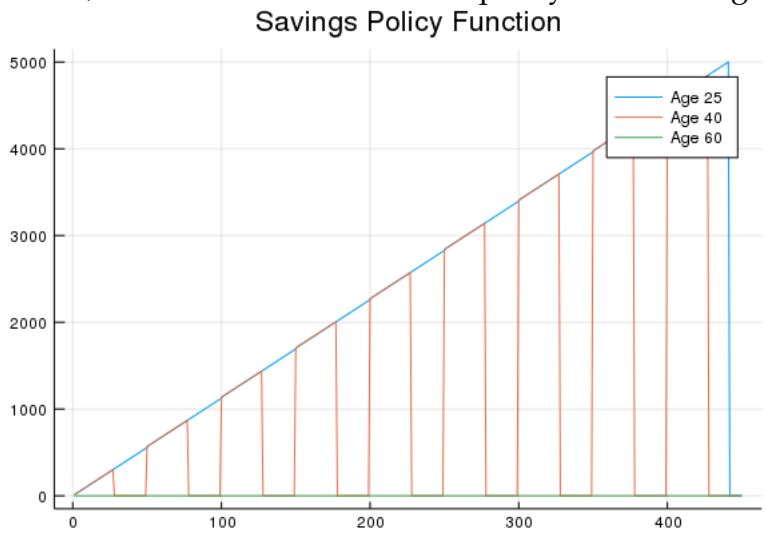
Homework 5

Alejandro Flores-Brown

03/06/2019

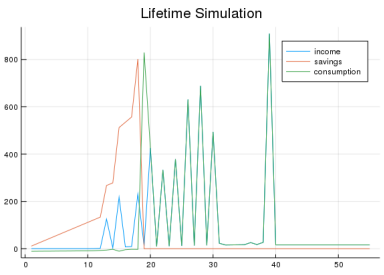
1 Problem 1

The given income process is negative for some ages, so we cannot simply take logs to transform income. Therefore I set income to be -4 when estimated income is negative. Otherwise, the code is as normal. The policy function is given below.



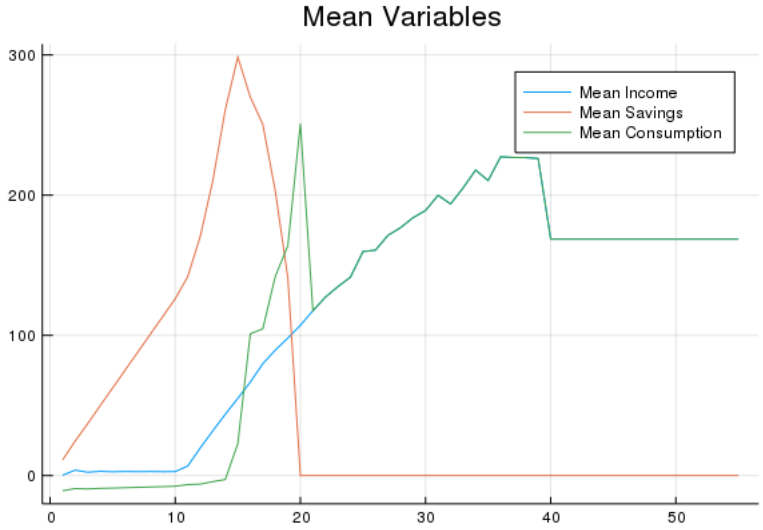
1.1 Part B

I now simulate the lifetime consumption, saving, and income for one individual.

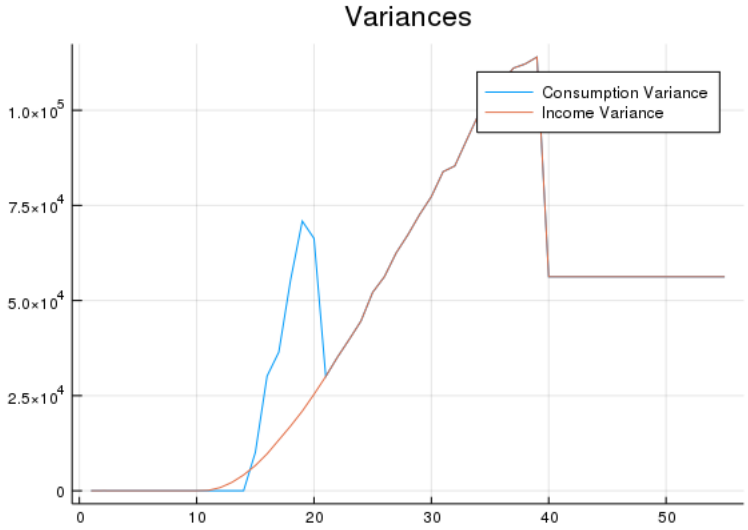


1.2 Part C

I next simulate the lifetime consumption, saving, and income for 10000 individuals.

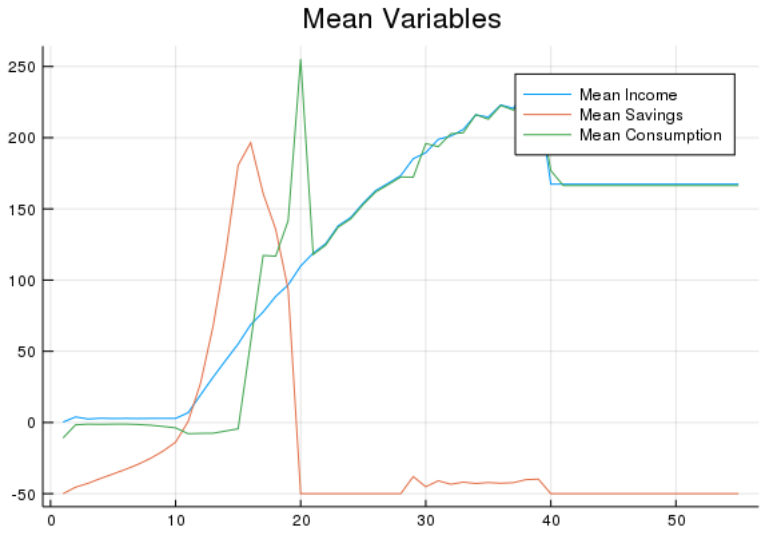


Variances are given below.

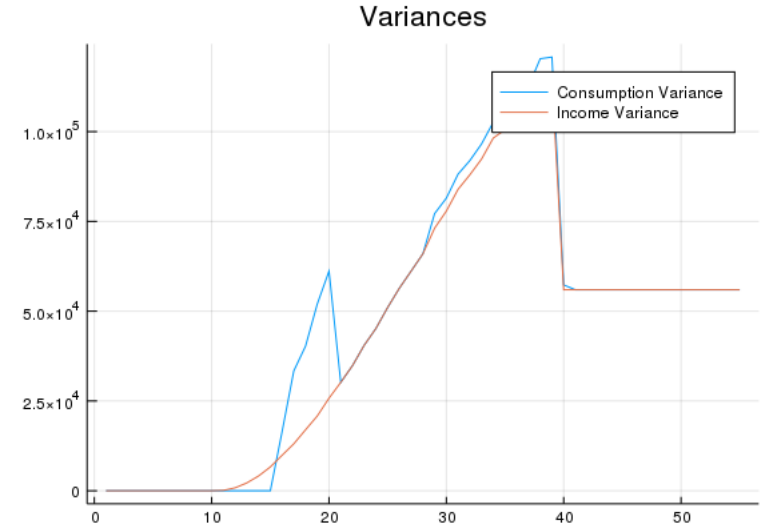


1.3 Part D

I next do the same simulation with a lower borrowing constraint.



Variances are given below.



Julia Code

```
using Plots, LinearAlgebra

n_l=25
n_h=80
mn = n_h    n_l

n_t=65
sigma=2
Gamma=0.92
r=0.02
R=1+r
z=[log(15); log(80); log(500)]
Gamma=[0.6 0.35 0.05;0.1 0.8 0.1;0.05 0.35 0.6]

f = zeros(mn)
for i=1:mn
    if 1.0135+0.1086*i 0.001122*i^2 > 0
        f[i] = log( 1.0135+0.1086*i 0.001122*i^2)
    else
        f[i] = 4
    end
end

epsilon = [log(0.8) log(1.2)]
pepsilon = [0.5 0.5]
exepsilon = dot(pepsilon,epsilon)
m1 = length(epsilon)

Piz = [1/3 1/3 1/3] * Gamma^1000
exz = dot(Piz,z)
m2 = length(z)

a_l = 0
a_h = 5000
```

```
m3 = 450

function calval(age::Int,income::Int,shock::Int,assets::Int,
    valfun::Array{Float64,4},a::Array{Float64,1})
     local Rz = 0
     local v = 0
     local s = 0

     if age+25>64
          Rz = 0.7*exp(z[income] + f[n_t    25])
          s=1
     end

     if age+25==80
          if R*a[assets] +Rz < 0
               v= (1/(1 sigma))*(R*a[assets]+Rz)^(1 sigma)
          else
               v =  999999 99*abs(R*a[assets]+Rz)
          end
     elseif age+25>65
          local V1 = zeros(m3)
          for i=1:m3
               if R*a[assets] + Rz    a[i]<0
                    V1[i] =  999999 99*abs(R*a[assets]+Rz a[i])
               else
                    V1[i]=(1/(1 sigma))*(R*a[assets]+Rz a[i])^(1
                       sigma)+Gamma*valfun[i,income,shock,age 1]
               end
          end
          helpv = findmax(V1)
          v = helpv[1]
          s = helpv[2]
     elseif age+25 == 65
          local V1 = zeros(m3)
          for i=1:m3
               if R*a[assets]+exp(z[income]+f[n_t 25]+epsilon[
                  shock]) a[i]<0
```

```
            V1[i] = -999999.99*abs(R*a[assets]+exp(z[income
                ]+f[age]+epsilon[shock]) - a[i])
        else
            V1[i]=(1/(1-sigma))*(R*a[assets]+exp(z[income]+
                f[age]+epsilon[shock]) - a[i])^(1-sigma)+Gamma
                *valfun[i,income,shock,age+1]
        end
    end
    v = maximum(V1)
elseif age+25 > 25
    local V1 = zeros(m3)
    for i=1:m3
        if R*a[assets]+exp(z[income]+f[n_t - age+25]+epsilon[
            shock]) - a[i]<0
            V1[i] = -999999.99*abs(R*a[assets]+exp(z[income
                ]+f[age]+epsilon[shock]) - a[i])
        else
            local n = n_t -  age +  24
            V1[i]=(1/(1-sigma))*(R*a[assets]+exp(z[income]+
                f[age]+epsilon[shock]) - a[i])^(1-sigma)
            +Gamma*(Gamma[income,1]*(0.5*valfun[i,1,1,n] +
                0.5*valfun[i,1,2,n]) + Gamma[income,2]*(0.5*
                valfun[i,2,1,n]+0.5*valfun[i,2,2,n])
            + Gamma[income,3]*(0.5*valfun[i,3,1,n]+0.5*
                valfun[i,3,2,n]) )
        end
    end
    helpv = findmax(V1)
    v = helpv[1]
    s = helpv[2]
else
    local V1 = zeros(m3)
    for i=1:m3
        if exp(z[income]+f[n_t - age+25]+epsilon[shock]) - a[i
            ]<0
            V1[i] = -999999.99*abs(exp(z[income]+f[age]+
                epsilon[shock]) - a[i])
```

```
            else
                local n = n_ret   age   24
                V1[i]=(1/(1 sigma))*(exp(z[income]+f[age]+
                    epsilon[shock]) a[i])^(1 sigma)
                +Gamma*(Gamma[income,1]*(0.5*valfun[i,1,1,n] +
                    0.5*valfun[i,1,2,n]) + Gamma[income,2]*(0.5*
                    valfun[i,2,1,n]+0.5*valfun[i,2,2,n])
                + Gamma[income,3]*(0.5*valfun[i,3,1,n]+0.5*
                    valfun[i,3,2,n]) )
            end
        end
        helpv = findmax(V1)
        v = helpv[1]
        s = helpv[2]
    end
    return v,s
end

function value(agrid::Array{Float64,1})
    local valfun=zeros(m3,m2,m1,mn)
    local s = zeros(m3,m2,m1,mn)
    for shock=1:m1
        for age=1:mn
            for income=1:m2
                for assets=1:m3
                    helpv = calval(age,income,shock,assets,
                        valfun,agrid)
                    valfun[assets,income,shock,age] = helpv[1]
                    s[assets,income,shock,age] = agrid[helpv
                        [2]]
                end
            end
        end
    end
    return valfun,s
end
```

```
function calcons(valfun::Array{Float64,4}, s::Array{Float64,4})
    local c = zeros(m3,m2,m1,mn)

    for assets=1:m3
        for income=1:m2
            for shock=1:m1
                local Rz = 0.7*exp(z[income] + f[n_t 25])
                c[assets,income,shock,n_h 25] = R*a[assets]+Rz
            end
        end
    end

    for age = 1:(n_h n_t 26)
        for assets=1:m3
            for income=1:m2
                for shock=1:m1
                    local Rz = 0.7*exp(z[income]+f[n_t 25])
                    c[assets,income,shock,n_t age 25] = R*a[
                        assets]+Rz s[assets,income,shock,n_h n_t
                         age 25]
                end
            end
        end
    end

    for assets=1:m3
        for income=1:m2
            for shock=1:m1
                c[assets,income,shock,n_t 25] = R*a[assets]+exp
                    (z[income]+f[n_t 25]+epsilon[shock]) s[
                    assets,income,shock,n_t 25]
            end
        end
    end

    for age=1:n_t 26
        for assets=1:m3
```

```
            for income=1:m2
                for shock=1:m1
                    c[assets,income,shock,n_t 25 age] = R*a[
                        assets]+exp(z[income]+f[n_t 25]+epsilon[
                        shock]) s[assets,income,shock,n_t age
                         25]
                end
            end
        end
    end

    for assets=1:m3
        for income=1:m2
            for shock=1:m1
                c[assets,income,shock,1] = exp(z[income]+f[1]+
                    epsilon[shock]) s[assets,income,shock,1]
            end
        end
    end
    return c
end

a1 = collect(range(0; length=450,stop=5000))
a2 = collect(range( 50; length=450,stop=5000))

VS1 = value(a1)
V1 = VS1[1]
S1 = VS1[2]
C1 = calcons(V1,S1)

a25 = zeros(450)
a40 = zeros(450)
a60 = zeros(450)

for i=1:450
    a25[i] = S1[i,1,1,1]
    a40[i] = S1[i,1,1,16]
```

```
    a60[i] = S1[i,1,1,36]
end

polfun = plot([a25 a40 a60],title="Savings Policy Function",
   label=["Age 25" "Age 40" "Age 60"])
savefig("f1")

#Part B

function sim(ntimes::Int,S::Array{Float64,4},a::Array{Float64
   ,1})

    cGamma = [0.6 0.95 1;0.1 0.9 1;0.05 0.4 1]
    za = zeros(n_h25,ntimes)
    iz = zeros(Int64,n_h25,ntimes)
    for i=1:ntimes
        za[1,i] = z[1]
        iz[1,i] = 1
    end

    pathi = zeros(n_h25,ntimes)
    for t=1:n_h25
        for i = 1:ntimes
            pathi[t,i] = rand()
        end
    end

    epsilona = zeros(n_h25,ntimes)
    iepsilon = zeros(Int64,n_h25,ntimes)
    paths = zeros(n_h25,ntimes)

    for i=1:ntimes
        for t=1:n_h25
            paths[t,i] = rand()
        end
    end
```

```
ysim = zeros(n_h25,ntimes)
csim = zeros(n_h25,ntimes)
asim = zeros(n_h25,ntimes)


for i=1:ntimes

    for t=2:n_t25
        if za[t1,i]==z[1]
            if pathi[t,i]<=cGamma[1,1]
                za[t,i]=z[1]
                iz[t,i] = 1
            elseif pathi[t,i]<=cGamma[2,1]
                za[t,i]=z[2]
                iz[t,i] = 2
            else
                za[t,i]=z[3]
                iz[t,i] = 3
            end
        elseif za[t1,i]==z[2]
            if pathi[t]<=cGamma[1,2]
                za[t,i]=z[1]
                iz[t,i] = 1
            elseif pathi[t,i]<=cGamma[2,2]
                za[t,i]=z[2,i]
                iz[t,i] = 2
            else
                za[t,i]=z[3]
                iz[t,i] = 3
            end
        else
            if pathi[t,i]<=cGamma[1,3]
                za[t,i]=z[1]
                iz[t,i] = 1
            elseif pathi[t,i]<=cGamma[2,3]
                za[t,i]=z[2]
                iz[t,i] = 2
```

```
            else
                za[t,i]=z[3]
                iz[t,i] = 3
            end
        end
    end

    for t=n_t 24:n_h 25
        za[t,i]=za[n_t 25,i]
        iz[t,i] =iz[n_t 25,i]
    end

    for t=1:n_h 25
        if paths[t,i]<=0.5
            epsilona[t,i] = epsilon[1]
            iepsilon[t,i] = 1
        else
            epsilona[t,i] = epsilon[2]
            iepsilon[t,i] = 2
        end
    end

    for t=1:n_t 26
        ysim[t,i]=exp(za[t,i]+f[t]+epsilona[t,i])
    end
    for t=n_t 25:n_h 25
        ysim[t,i]=0.7*exp(za[n_t 25,i]+f[n_t 25])
    end

    asim[1,i] = S[1,iz[1,i],iepsilon[1,i],1]
    csim[1,i] = ysim[1,i]   S1[1,iz[1,i],iepsilon[1,i],1]
    for t=2:n_h 25
        asim[t,i] = S[findfirst(x  > x==asim[t 1,i],a),iz[t
           ,i],iepsilon[t,i],t]
        csim[t,i] = R*asim[t 1,i]+ysim[t,i] asim[t,i]
    end
end
```

```
    return ysim,asim,csim

end

sm1 = sim(1,S1)


ages=[n_l:n_h 1]
sim1 = plot([sm1[1] sm1[2] sm1[3]],title="Lifetime Simulation",
   label=["income" "savings" "consumption"])
savefig("f2")

#Part C

sm2 = sim(10000,S1,a1)

function calstats(ymat::Array{Float64,2},smat::Array{Float64
   ,2},cmat::Array{Float64,2})
    ybar = zeros(n_h 25)
    sbar = zeros(n_h 25)
    cbar = zeros(n_h 25)

    for j=1:n_h 25
        for i=1:10000
            ybar[j] = ybar[j] + ymat[j,i]
            sbar[j] = sbar[j] + smat[j,i]
            cbar[j] = cbar[j] + cmat[j,i]
            lo
        end
        ybar[j] = ybar[j]/10000
        sbar[j] = sbar[j]/10000
        cbar[j] = cbar[j]/10000
    end

    return ybar, sbar, cbar
```

```
end

mS2 = calstats(sm2[1],sm2[2],sm2[3])

sim2 = plot([mS2[1] mS2[2] mS2[3]], title="Mean Variables",
   label=["Mean Income" "Mean Savings" "Mean Consumption"])
savefig("f3")

#Part D

VS2 = value(a2)
V2 = VS2[1]
S2 = VS2[2]
C2 = calcons(V1,S1)

sm3 = sim(10000,S2,a2)


mS3 = calstats(sm3[1],sm3[2],sm3[3])

sim3 = plot([mS3[1] mS3[2] mS3[3]], title="Mean Variables",
   label=["Mean Income" "Mean Savings" "Mean Consumption"])
savefig("f4")
```